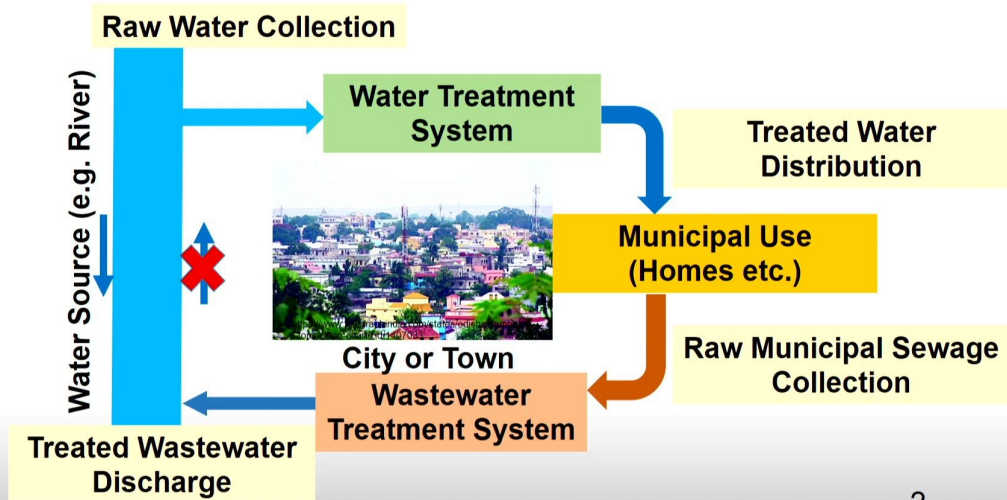




02:58

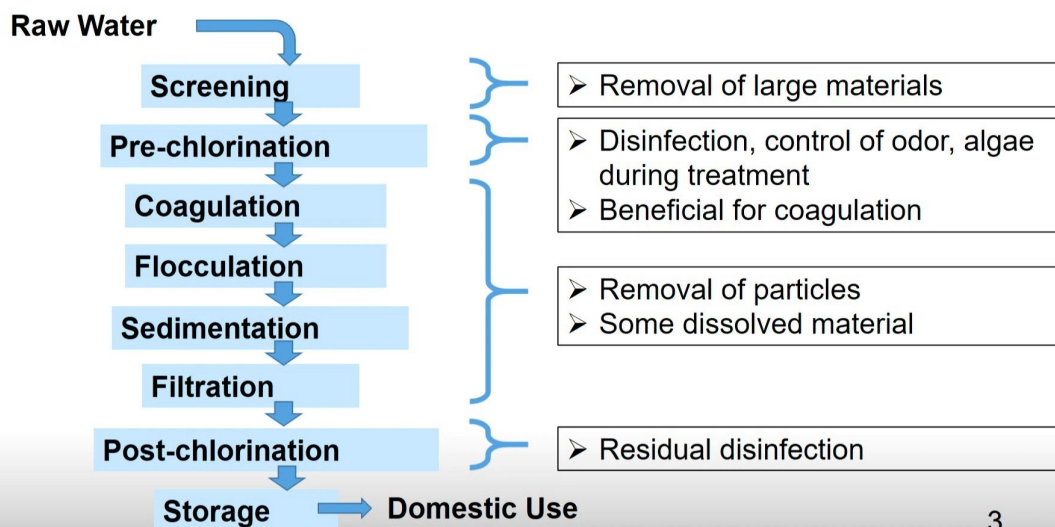
Typical Water Use Pattern



2

04:00

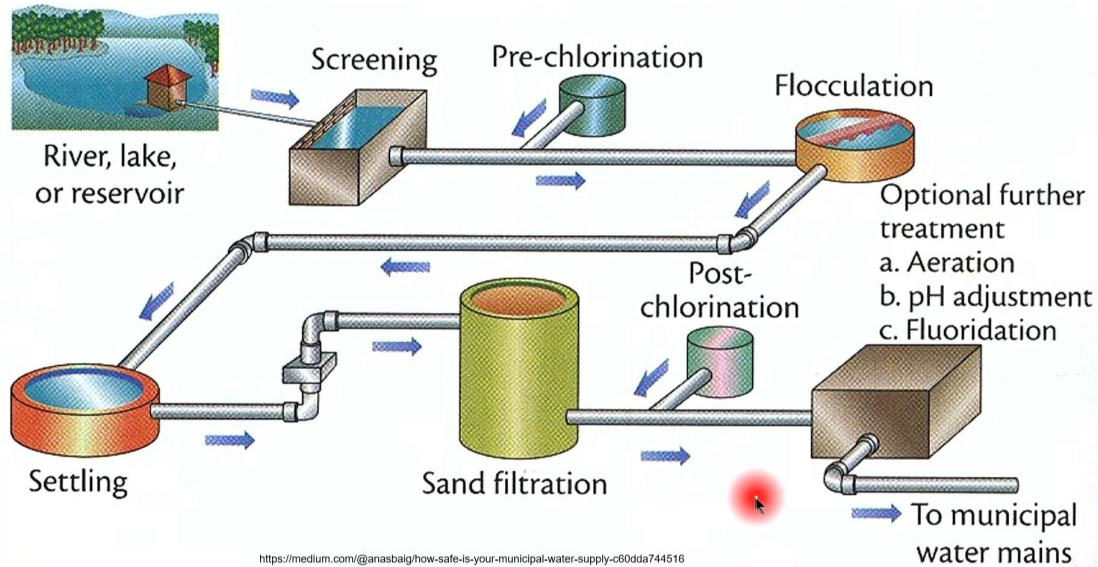
Municipal Surface Water Treatment



3

04:50

Municipal Surface Water Treatment: Treatment train



05:04

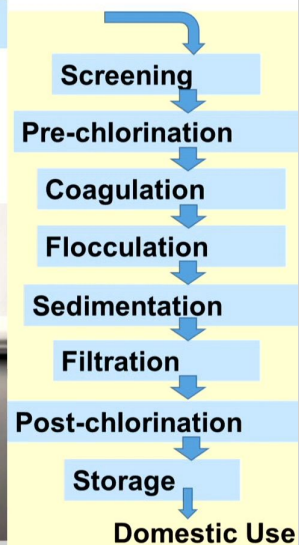
Municipal Surface Water Treatment

Coagulation – Flocculation – Sedimentation

- Once sufficiently large flocs are formed, they are allowed to settle by gravity. The process is called as **sedimentation or settling**.



<http://www.ecologixsystems.com/images/chemical-jar-tests.jpg>

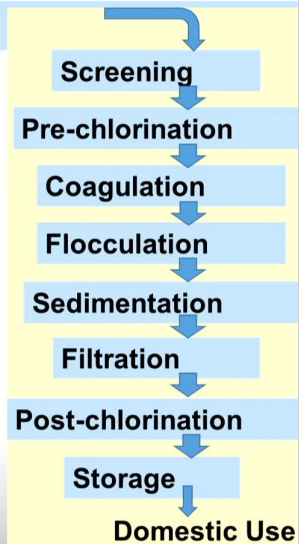
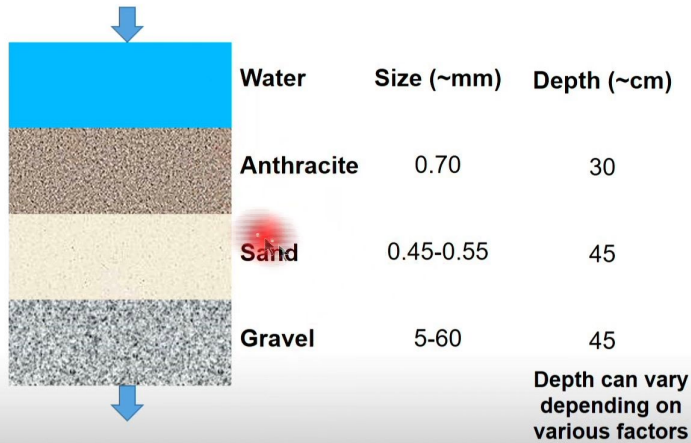


05:40

Municipal Surface Water Treatment

Filtration: Rapid Sand Filter

❑ Some flocs still resist settling



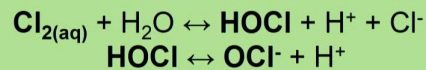
06:51

Municipal Surface Water Treatment

Disinfection by Chlorination

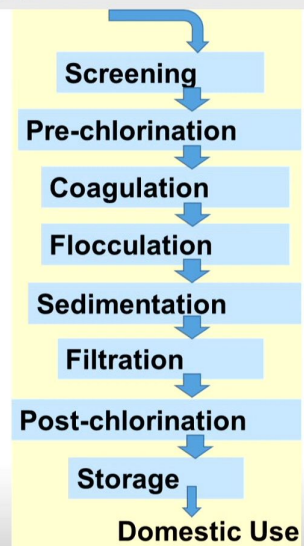
- **Primary disinfection:** To kill any pathogens in the water
- **Secondary (or Residual) disinfection:** To prevent pathogen regrowth in the water during the period before use

Free Chlorine Disinfection



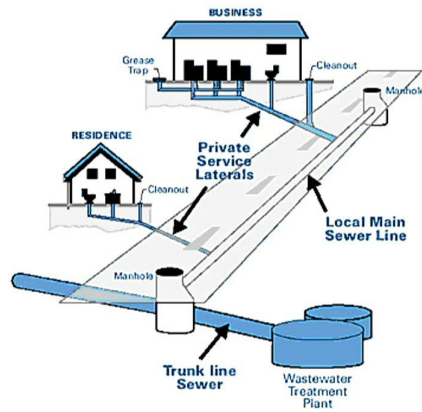
HOCl : Hypochlorous acid
OCl⁻ : Hypochlorite Ion

Disinfection by-products: Carcinogenic in nature



08:55

Municipal Wastewater Collection and Transport



<https://www.conshohockensa.com/media/3358/Sewer-System-Diagram.jpg>

Arceivala and Asolekar, 2007

- ❖ Critical for centralized treatment
- ❖ Usually sewerage system costs account for about 80% of the total cost, while treatment may account for only 20% of the total cost.
- ❖ Gravity flow
- ❖ Several meter down when reach to treatment plants

9

09:27

Conventional Municipal Wastewater Treatment

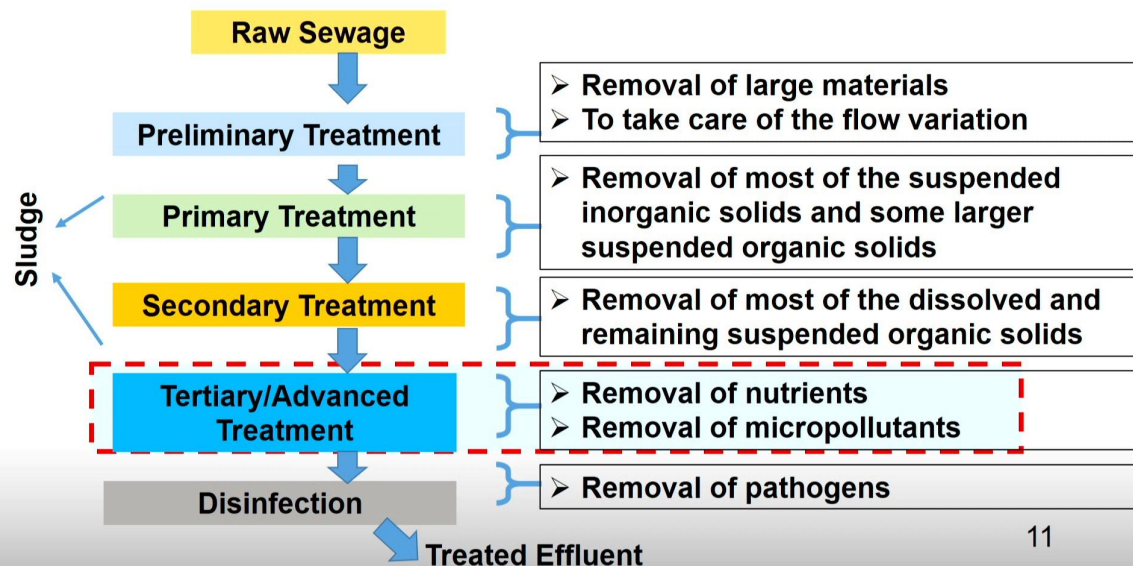
❑ Composition of Municipal Wastewater (Sewage)

- ❖ Large solid objects (animal/vegetable matter, paper, plastic etc.)
- ❖ Inorganic solids (sand, silt, clay): (a) > 0.2mm (Grit), (b) Suspended Solids, (c) Dissolve Solids
- ❖ Organic solids: (a) SS, (b) DS (may include nutrients, oil, grease etc)
- ❖ Nutrients
- ❖ Pathogens
- ❖ Micropollutants

10

10:56

Conventional Municipal Wastewater Treatment



[11:41](#)

Conventional Municipal Wastewater Treatment

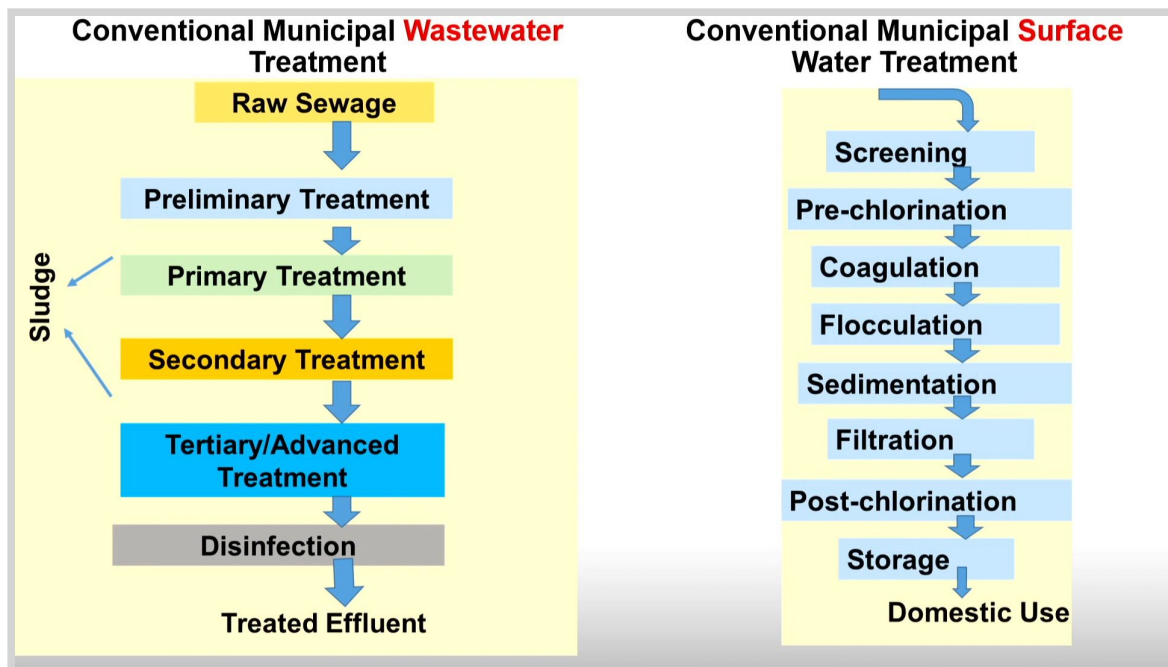
Tertiary Treatment

❑ After Secondary Treatment, WW still contains:

- ❖ Dissolved inorganic matter
- ❖ Nutrients (N, P)
- ❖ Micropollutants
- ❖ Pathogens

12

[12:52](#)



13:34

Why advance treatment?

- ❖ Conventional treatment systems failed to remove some emerging pollutants

Major Pollutants in Water

❑ Three major category

- ❑ Inorganic Pollutants
- ❑ Organic Pollutants
- ❑ Biological Pollutants



- ❑ Nutrients: N, P; Heavy Metals, Fluoride etc.
- ❑ Pesticides, PPCPs etc.
- ❑ Pathogens

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14:06

Major Pollutants in Water

Organic Pollutants

- ❖ Persistent Organic Pollutants (POPs)
- ❖ Polycyclic Aromatic Hydrocarbons (PAHs)
- ❖ **Emerging Contaminants: Pharmaceutical and Personal Care Products (PPCPs)**

Persistent organic pollutants (POPs)

- ❑ Persistent organic pollutants (POPs) are organic compounds that are resistant to environmental degradation through **chemical, biological, and photolytic processes**
- ❑ **Stockholm Convention on Persistent Organic Pollutants in 2001**

15

[15:13](#)

Major Pollutants in Water

❑ Emerging Contaminants: Pharmaceutical and Personal Care Products (PPCPs)

- ❖ **A unique group of emerging environmental contaminants**, due to their inherent ability to **induce physiological effects in humans and others organisms at low doses.**

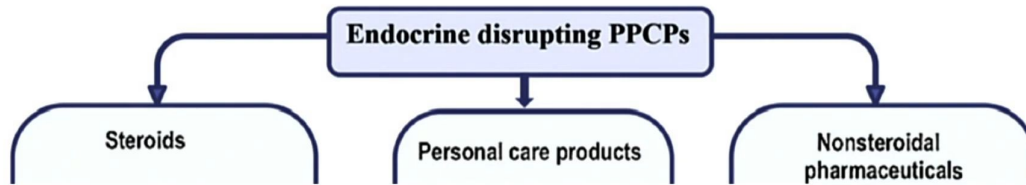
- ❑ Drugs
- ❑ Fire retardants
- ❑ Disinfectants
- ❑ Fragrances
- ❑ Insecticides/ Repellants

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[16:00](#)

❑ Emerging Contaminants: Pharmaceutical and Personal Care Products (PPCPs)

Harmful effects?



❑ Endocrine disrupting chemicals

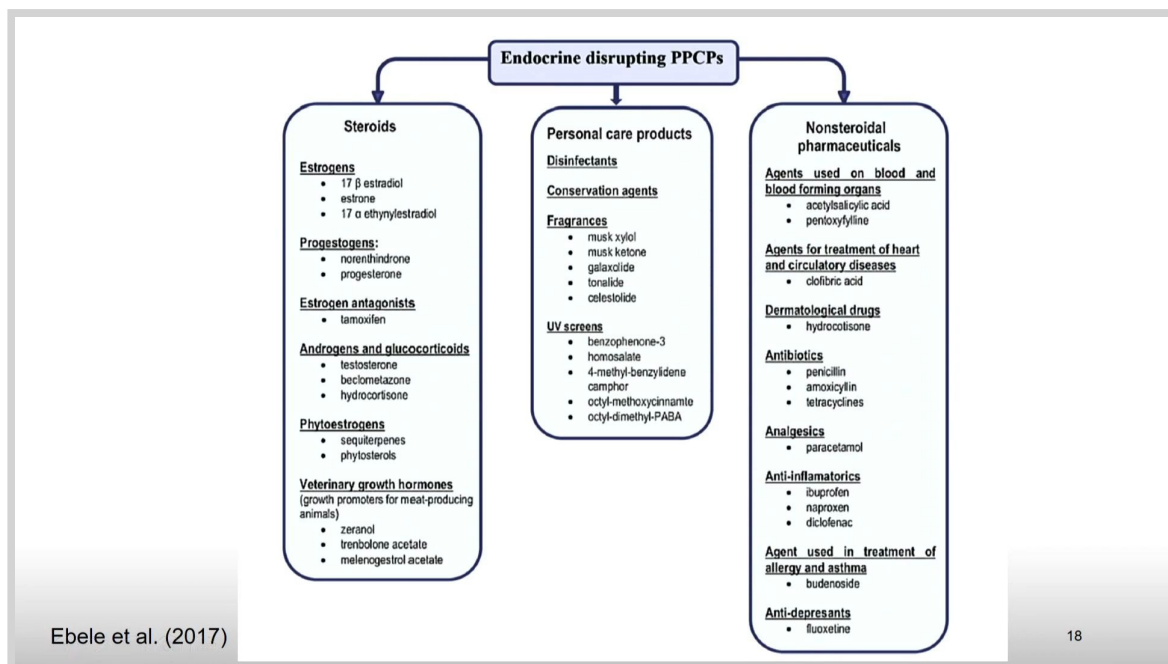
❖ An chemical agent that interferes with **synthesis, secretion, transport, metabolism, binding action, or elimination of natural blood-borne hormones** that are present in the body and are responsible for homeostasis, reproduction, and developmental process

❑ **Homeostasis** refers to stability, balance, or equilibrium within a cell or the body. It is an organism's ability to keep a constant internal environment.

Ebele et al. (2017)

17

18:13

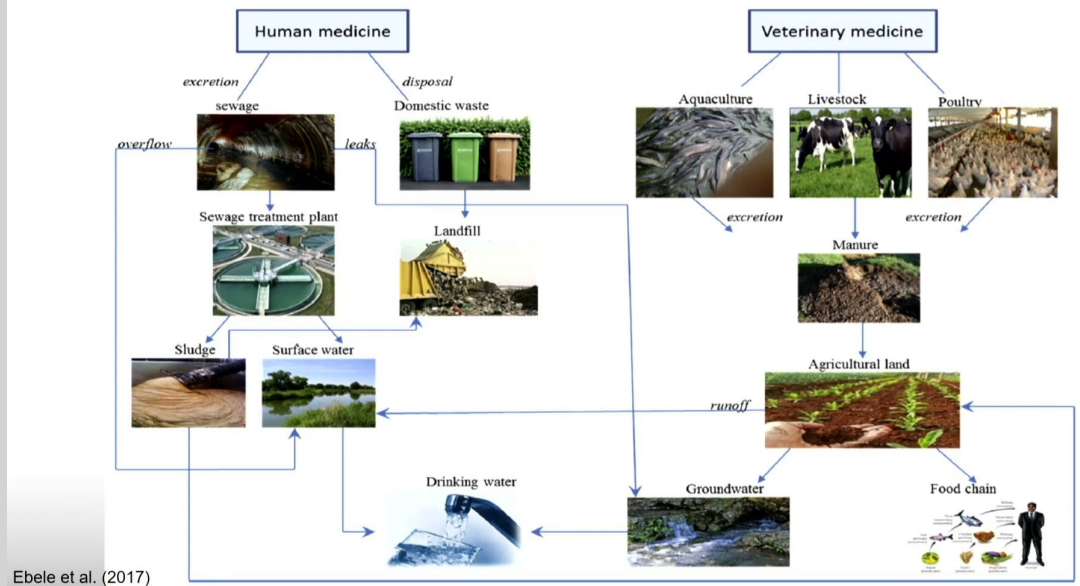


Ebele et al. (2017)

18

19:34

How PPCPs (drugs) reach to water?



23:06

Advance Treatment Technologies

❖ Advance Oxidation Processes (AOP's)

- Photooxidation
- Ozonation
- Fenton process
- Electrooxidation

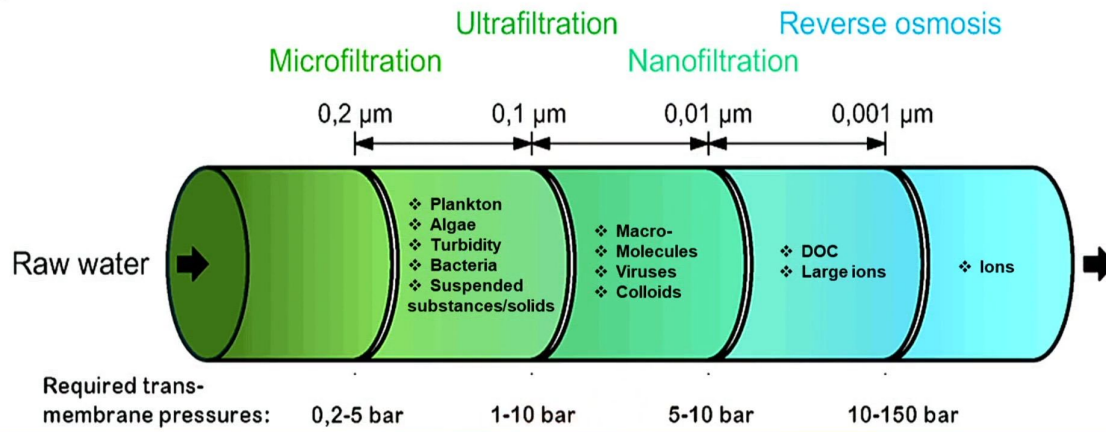
❖ Membrane Processes

❖ Photobioreactors

20

27:33

Background: membrane technology



❑ A membrane is a **selective barrier**; it allows **some things to pass through but stops others**. Such things may be molecules, ions, or other small particles

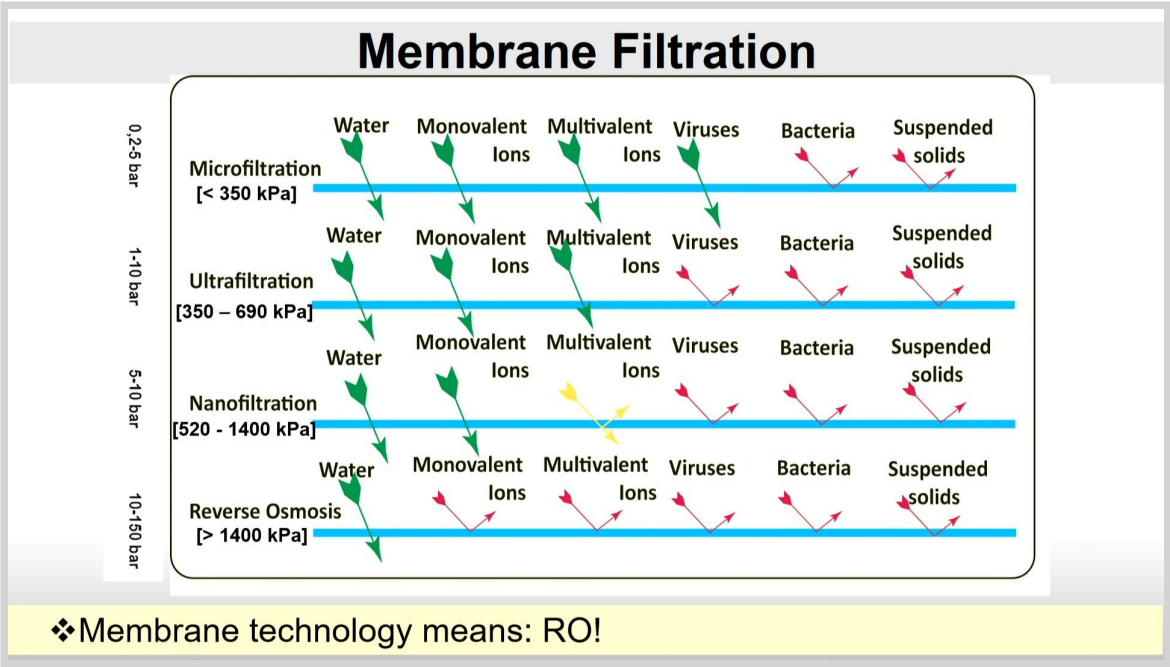
29:03

Background: membrane technology

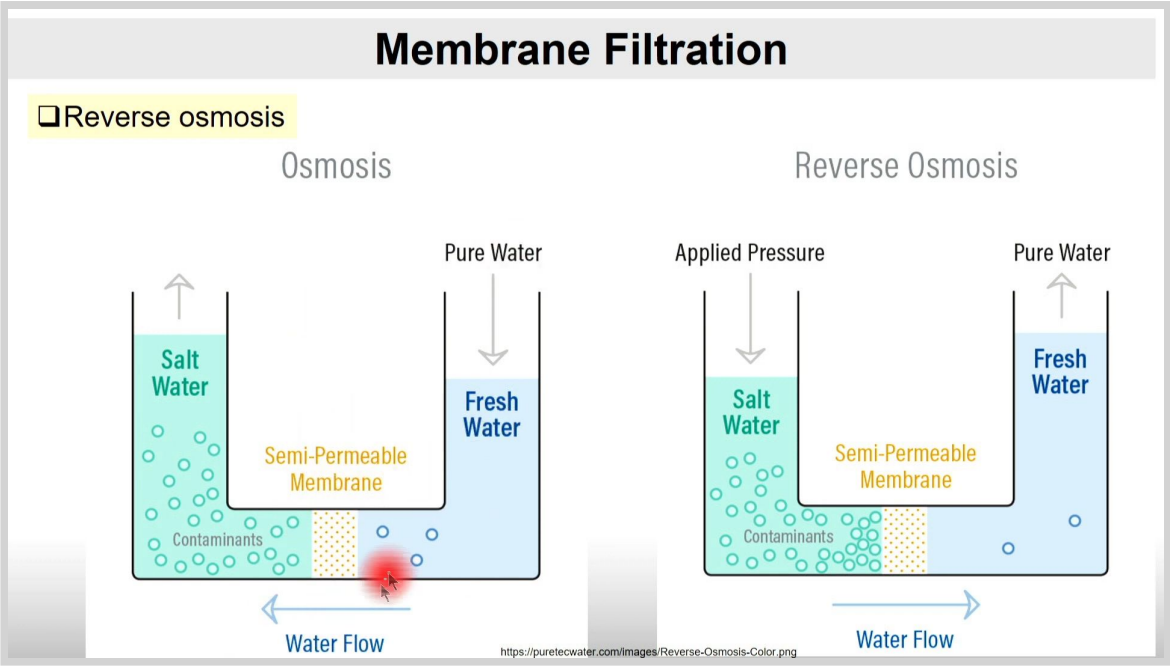
- ❖ Started ~1960 for water & wastewater treatment.
- ❖ Challenges due to high energy requirement and lack of membrane materials.
- ❖ Extensive use after 1990, materials innovation, energy recovery units and strict water quality discharge.
- ❖ Water reuse annual growth rate is 14%, and desalination is 8%. [1,2,3]
- ❖ Annual growth rate for membrane filtration is ~17%. [1,2]
- ❖ **Lower footprint(area)**

[1] Global Water Intelligence Wastewater Reuse Report, 2005
 [2] Global Water Market 2008: Opportunities in Scarcity and Environmental Regulation Global Water Intelligence, October 2007
 [3] Global Markets and Technologies for Water Recycling and Reuse, 2017
<https://www.hydrogroup.biz/areas-of-use/water-treatment/membrane-filtration.html>

30:16

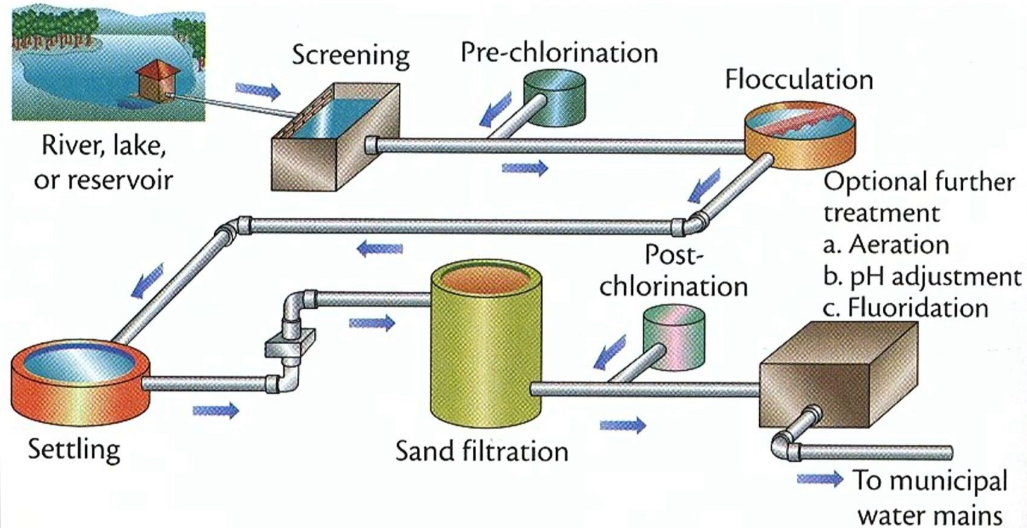


30:37



31:08

Municipal Surface Water Treatment: Treatment train

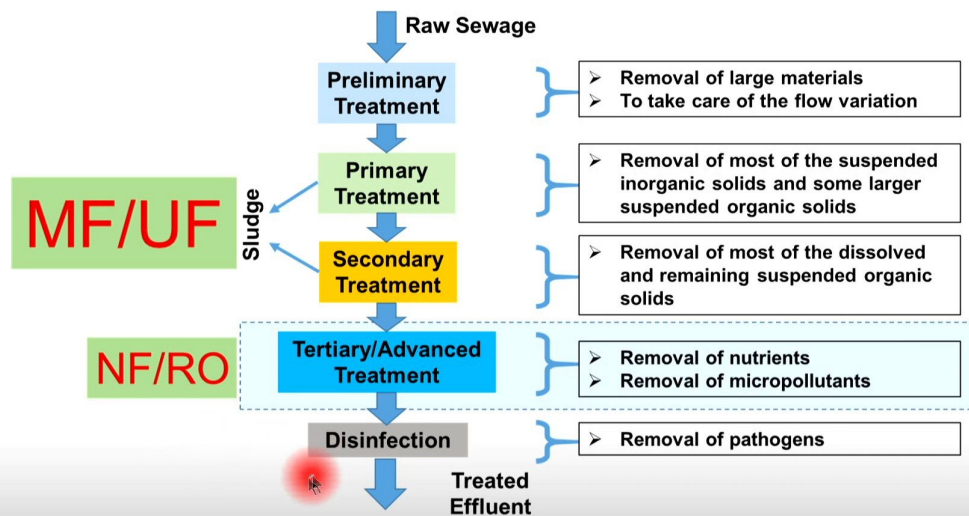


<https://medium.com/@anasbaig/how-safe-is-your-municipal-water-supply-c60dda744516>

25

[32:59](#)

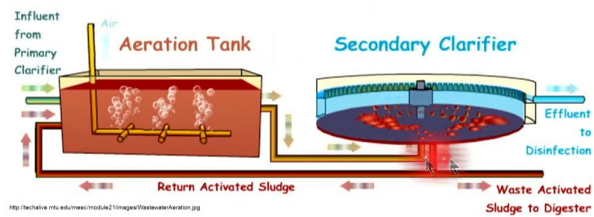
Municipal wastewater Treatment: Treatment train



26

[33:26](#)

Membrane bioreactor (MBR)



27

33:37

Membrane bioreactor (MBR)

- ❖ **Membrane bioreactor (MBR)** is the combination of a membrane process like microfiltration or ultrafiltration with a biological wastewater treatment process, the **activated sludge process**.

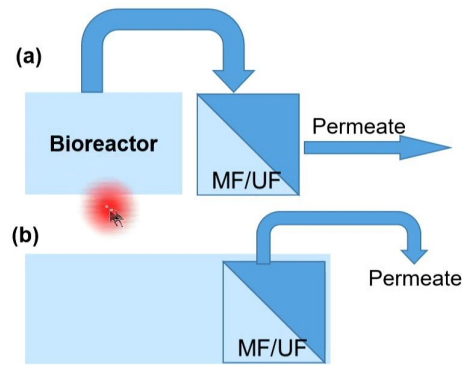


27

34:23

Membrane Bioreactor (MBR)

- ❑ Integrated model ASP & membrane separation process
- ❑ High effluent quality
- ❑ **Small footprint**
- ❑ Low-pressure membranes (UF/MF) used
- ❑ Side stream and submerged configuration
 - ❑ Each have their own advantages and disadvantages
 - ❑ Submerged is the most common due to less energy consumption, compact design



Arrangements in MBR, (a) Side-stream membrane, (b) Submerged membrane

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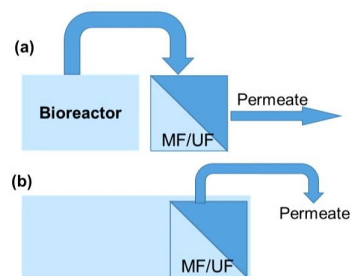
35:23

Membrane Bioreactor (MBR)

❑ Problems in MBR

- ❖ Cannot remove poorly biodegradable micropollutants.
 - Less removal efficiency achieved for pharmaceutical compounds.
- ❖ Higher energy consumption due to membrane fouling

- ❑ **Hybrid processes** like MBR-NF/RO, osmotic membrane bioreactor (OMBR), Membrane distillation bioreactor (MDBR) significantly decrease the quantity of micropollutants from wastewater (Luo et al., 2017; Chon et al., 2011)



Arrangements in MBR, (a) Side-stream membrane, (b) Submerged membrane

Chon, K., Sarp, S., Lee, S., Lee, J. H., Lopez-Ramirez, J. A., & Cho, J. (2011). *Desalination*, 272(1–3), 128–134. <https://doi.org/10.1016/j.desal.2011.01.002>
 Luo, W., Phan, H. V., Xie, M., Hai, F. I., Price, W. E., Elimelech, M., & Nghiem, L. D. (2017). *Water Research*, 109, 122–134. <https://doi.org/10.1016/j.watres.2016.11.036>

29

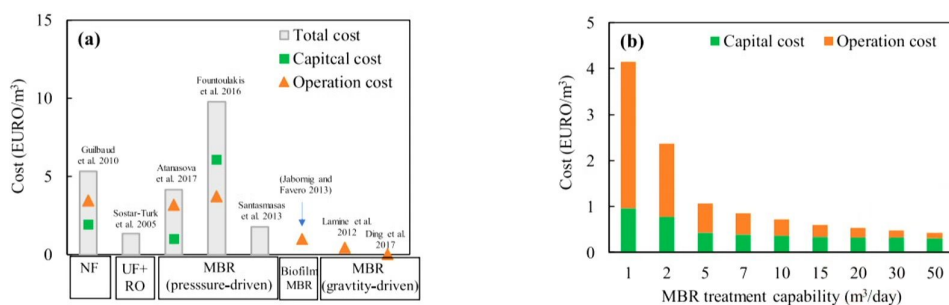
36:44

Current status: Membrane-based wastewater Treatment

- ❖ Well recognized as an alternative water resource for non-potable or potable use.
- ❖ Membrane-based techniques with superior treated water quality.
- ❖ The membrane-based processes:
 - Membrane filtration, hybrid membrane systems, and resource recovery oriented membrane based systems for wastewater treatment (non-potable and potable use).
 - Resource recovery is limited, however the concentrated retentate from MBR or membrane filtration can be used.
- ❖ Hybrid membrane systems such as UF-NF; UF-RO; MBR-NF have shown their potential.

37:54

Summery: Cost Estimation of Membrane-based Greywater Treatment



(a) A summary of cost estimation of membrane-based greywater treatment processes reported in previous studies; (b) Cost of MBR-based greywater treatment at different treatment capabilities, data are adopted from **Atanasova et al. (2017)**

- ❖ Future research: Nutrients removal and recovery, micropollutants removal, and life cycle assessment in membrane based greywater treatment needed to be systematically investigated

Adopted from <https://www.sciencedirect.com/science/article/pii/S004896971834703X> (Bing Wu, 2019)

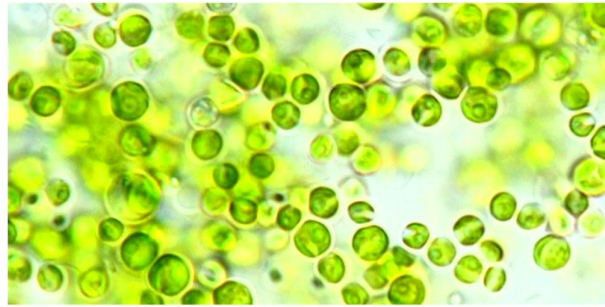
Atanasova, N., Dalnau, M., Comas, J., Poch, M., Rodriguez-Roda, I., Buttiglieri, G., 2017. Optimized MBR for greywater reuse systems in hotel facilities. J. Environ. Manag. 193, 503–511

31

39:25

Algae Based Treatment Systems

- ❖ Algae produce oxygen during photosynthesis which can meet the aeration requirements in conventional treatment.
- ❖ Symbiotic relation between aerobic bacteria and algae could contribute in organic carbon as well as nutrients removal.
- ❖ Algae also have potential to uptake various micropollutants.
- ❖ The generated algal biomass has huge commercial value.

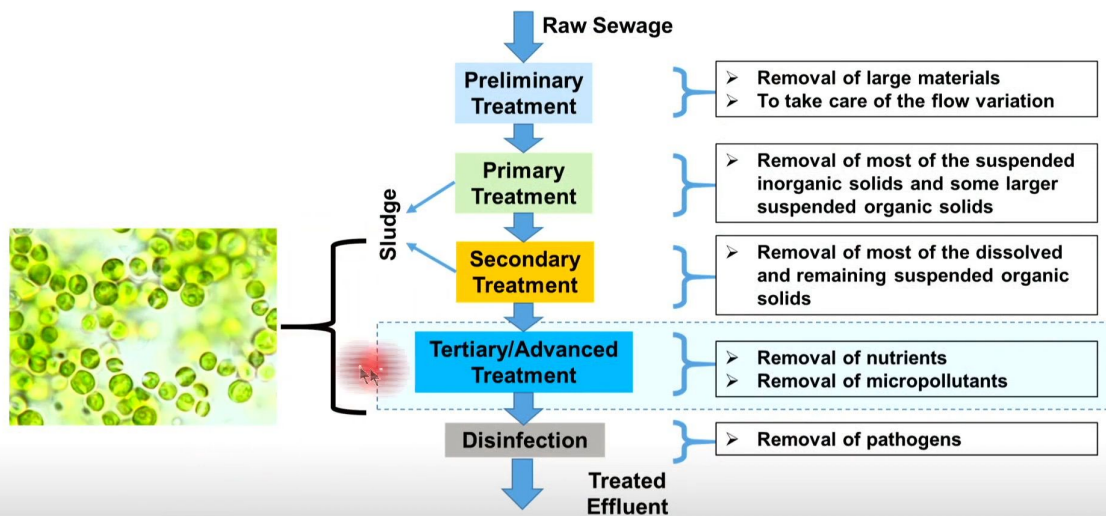


<http://www.valuefood.info/wp-content/uploads/2013/12/health-benefits-of-chlorella-670x337.jpg>

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40:05

Municipal wastewater Treatment: Treatment train



33

40:22

Algae Based Treatment Systems

Photobioreactor



34

40:50

Algae Based Treatment Systems

Photobioreactor



35

42:39

Wastewater Treatment in Space



- ❑ 1 liter bottle of water: ~ **\$22,000(~17.5 Lacs)** in International Space Station(ISS).
- ❑ Supporting a crew of six astronauts on the ISS requires about 6,800 liters of water to be launched per year at a cost of almost \$150 million!

- ❑ 100% recycling of water ISS(USA-NASA)
- ❑ Only 74% efficiency
- ❑ Need to Improve for long mission (Mars and beyond)

❑ ISRO also looking for better technology for the long mission

[43:13](#)

ES-200 Questions

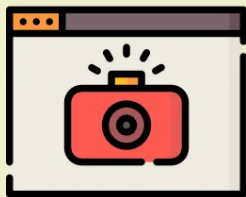
- ❑ Treatment train for conventional surface water treatment.
- ❑ Treatment train for conventional municipal wastewater treatment.
- ❑ Reasons for membrane technology higher growth rate.
- ❑ Activated sludge process vs. MBR.
- ❑ Why is RO technology not suitable for homes in India?
- ❑ Advantages with Algal Photobioreactor.
- ❑ Why are space agencies looking for high efficiency for wastewater recycling?

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